Reliability Analysis of CMOS Inverter Subjected to AC & DC NBTI Stresses

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Abstract— **This paper presents an experimental analysis of the impact of AC- and DC-type Negative Bias Temperature Instability (NBTI) stresses on the CMOS inverter DC response and robustness. The results reveal, on one side, that the inverter DC response under AC NBTI presents a parallel shift of that shown under DC NBTI. However, the AC- to DC-induced shift of the inverter logic threshold is found much less than ½ expected in the literature for a 0.5 duty-cycle AC NBTI. Furthermore, valid input logic levels are found much less affected by AC NBTI than by DC NBTI. On the other side, the inverter noise margins have followed different trends under AC NBTI compared to DC NBTI. In fact, while noise margin high (NMH) has increased under both DC- & AC-type NBTI, noise margin low (NML) has shrunk by DC NBTI and trivially affected by AC NBTI. Indeed, the inverter robustness is much less degraded by AC-type than by DC-type NBTI. And as such, NBTI models driven from DC stress may overestimate the degradation and therefore improperly predicts the circuit lifetime subjected to NBTI.**

Keywords-component: IC Reliability, CMOS Inverter, AC NBTI, DC NBTI, Noise Margins, IC Robustness.

I. INTRODUCTION

Negative bias temperature instability, a degradation phenomenon affecting mainly PMOS transistors [1-2], is becoming a major reliability issue for deep submicron CMOS integrated circuits. It is well known that this phenomenon is ascribed to Si/SiO_2 interface states and positive charges in the oxide resulting from the breaking down of Si-H bonds at the SiO_2/Si interface of PMOS transistors under negative bias at elevated temperature. NBTI is characterized by a positive shift in the absolute value of the PMOS threshold voltage, a decrease of the transistor's transconductance g_m and the drain current I_{DS}. Accordingly, circuit performances are degraded [3-5] and their lifetime is thereby shrunk. The amount of the degradation depends essentially on the stress time, the negative bias, and the temperature conditions under which the circuit is operating.

To predict circuit lifetime, a designer usually conducts an aging simulation where the circuit is first simulated using fresh models of the PMOS transistors, then the simulation is redone using aged models of these transistors. The aged models might monitor NBTI impact on PMOS v_{th0} parameter [6] or on both v_{th0} and μ_0 parameters [7]. Most of NBTI models used in industrial reliability simulators are extracted from device level characterization of DC NBTI. Nevertheless, the focus on the low-level details does not provide insight about the circuit degradation due to NBTI effect on a larger scale. This is due to the fact that these device-level models mostly focus on the degradation of a single transistor. However, they don't take into account the arrangement of PMOS transistors within a circuit, the device geometries, and most importantly, the actual stimuli applied to the circuit during operation and which strongly determine the magnitude and the speed of the degradation. So, to properly evaluate NBTI degradation, a circuit level characterization is required to adjust device-level model-driven aging simulation. In this prospect, we have characterized NBTI at circuit level by investigating its impact on CMOS inverter DC performances subjected to both DC- and AC-type stresses. In the experiments, a series of negative voltages have been applied at room temperature and at elevated temperature for a stress period of 3600s. For the AC-type NBTI, a 0.5 duty cycle AC pulse signal of 10 kHz frequency was applied. The Measure/Stress/Measure (MSM) technique was adopted for the test protocol.

The rest of the paper is organized as follows. In section II, the samples, NBTI experimental setup details, and MSM protocol are described. The obtained results are analysed and discussed in section III and section IV concludes the paper.

II. EXPERIMENTAL SETUP

The circuit under test (CUT) are CMOS inverters having 1µm/0.8µm and 2µm/0.8µm as aspect ratio for NMOS and PMOS respectively. The CUTs belong to the same wafer fabricated at ISiT (Institute for Silicon Technology) of Fraunhofer, Germany using 1µm CMOS process technology with 20 nm SiO_2 as gate oxide. This relatively thick oxide does not affect the study as NBTI degradation depends on the electric filed and not on the oxide thickness. It has been shown that devices of different gate oxide thickness T_{ox} show the same parameter degradation at the same oxide field E_{ox}. In fact, Gregor Pobegen et al. have presented in [8] a detailed study on equivalent devices with different oxide thicknesses (5 to 30 nm) where they have shown experimentally that the basic mechanisms behind the NBT instability are the same in thin and thick oxide technologies.

Our experimental setup is based on Measure/Stress/ Measure technique, where the sequences have been performed using fully automated benches. The bench includes a sensitive Agilent HP 4156C for voltage measurements and trigging, a

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Karl Suss PA300 micro-manipulator probe station, and a hotplate to vary the temperature of the chip during the experiments. The test circuit chip (bare-die) within the probe station were isolated from vibration and enclosed in a grounded faraday cage to avoid both RF and light effects.

The test protocol we adopted is described as follows: during stress period of 3600s, a DC negative voltage was applied to the inverter input for DC-type stress while an AC pulse signal of 0.5 duty-cycle and 10 kHz frequency was applied for AC-type stress. The rest of circuit pads were grounded in both types (as shown in fig. 1).

During the measurement, a conventional DC configuration was applied, where a DC voltage source of 5V was applied to the inverter's input and the output was monitored (measured). The switch between stress and measure configurations was automated via a LabVIEW program.

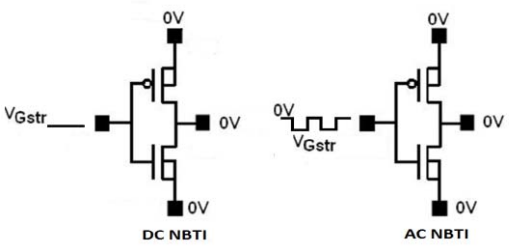


Fig.1 CMOS inverter configuration under DC&AC NBTI stresses

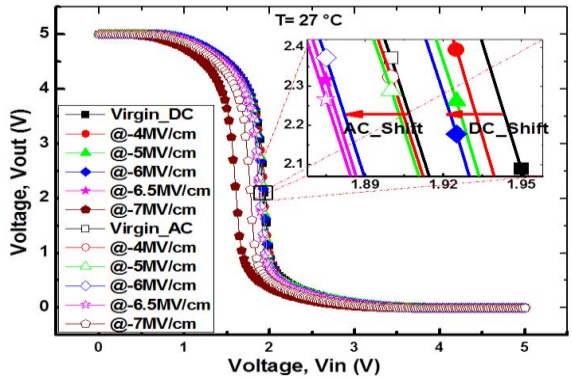
First, a measure at room temperature of the virgin (unstressed) inverter's voltage transfer curve (VTC) has been done and saved. Then, the inverter undergoes a DC (AC) stress for a period of 3600s. For NBTI DC stress, a negative voltage of -8V,-10V,-13 and -14V was applied to the inverter's input inferring a negative field of -4MV/cm to -7MV/cm. For NBTI AC stress, a pulse signal with the same magnitude of -8 to -14V was applied. After each stress, a measure of the inverter's VTC is done and its data have been saved. This test protocol has been repeated for elevated temperatures from the range of 80°C to 140°C with a 20°C step using a PID- controlled hotplate in order to combine large field with elevated temperature. As we are interested in analyzing and comparing the effect induced by AC NBTI stress against that induced by DC NBTI stress on the inverter DC features, then some results of the characterization of the inverter subjected to DC NBTI [9] are reproduced here to more perceive the two NBTI types effects.

III. RESULTS AND DISCUSSION

Fig. 2 shows the inverter`s VTC shift due to both AC and DC NBTI at room temperature in (a) and at elevated temperature in (b). Solid Symbols correspond to DC NBTI whereas open Symbols correspond to the AC NBTI. Virgin curves denote VTCs of unstressed inverters and the other curves correspond to the VTCs of the stressed ones with a series of electric field ranging from -4 to -7MV/cm.

It clearly outcomes that DC NBTI stress causes a VTC shift (with respect to unstressed VTC) larger than does AC NBTI stress at both room temperature and at elevated temperature but with a higher magnitude at this later. This is due to the fact that

(a)



(b)

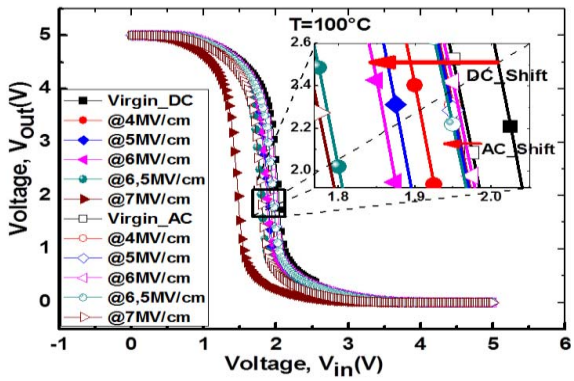


Fig.2 CMOS inverter VTC shift subjected to DC&AC NBTI stresses at room temperature in (a) and at elevated temperature in (b).

while the inverter undergoes just electric field acceleration at room temperature, it also undergoes at 100°C, a temperature acceleration stress. This later exacerbates more its degradation under DC NBTI stress.

It is worth to note that DC NBTI-induced shift of the inverter`s VTC increases with negative electric field in (a) and with both electric field and temperature in (b). However, AC NBTI- induced VTC`s shift does increase with field in (a) but not, so much, with both of them at high temperature (b). This can be due to the fact that temperature can also accelerate the recovery during the AC stress. This will be more discussed later.

To more apprehend the difference between AC and DC NBTI stresses on the CMOS inverter VTC, we have tracked their impact on the main critical voltages of the VTC. These voltages correspond to the inverter`s logic (switching) threshold voltage, valid input and output logic levels and noise margins which characterize the inverter circuit robustness. These voltages are extracted from the VTC as explained in [9]. Their respective trends under both NBTI stress types are shown in the following sub sections.

A. Inverter logic threshold under AC- & DC-type NBTI

Fig.3 shows the shift of the CMOS inverter logic (switching) threshold voltage under both DC- & AC-type NBTI in semi-log scale with respect to the energy in (a) and to the negative field in (b). The solid symbols represent experimental data of the shift due to DC NBTI while open symbols represent those due to AC-type stress. In the same way, the solid and dashed lines are the linear fit of DC and AC data respectively.

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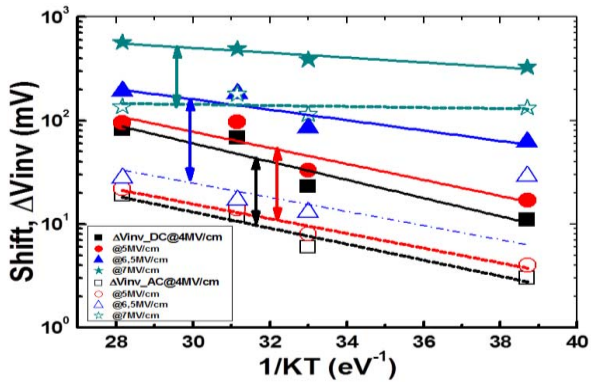
(a)



(b)

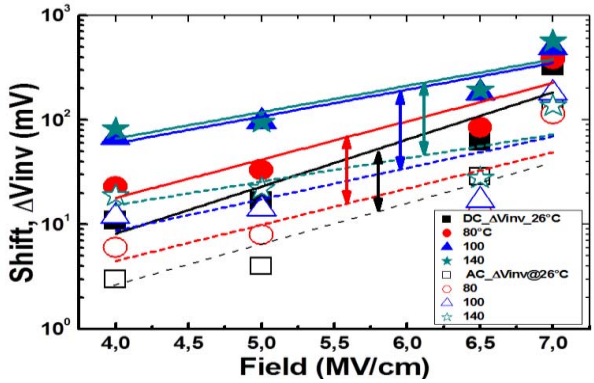


Fig.3 Inverter switching threshold shift under DC-& AC-type NBTI stresses with respect to energy (a) and the applied field (b).

The results show that the inverter logic threshold shift has undergone under AC NBTI a quasi-parallel shift of that under DC stress but with a lower degradation magnitude. Indeed, the AC slopes are slightly lower, within a range of 1 %, than DC slopes.

By observing the slopes of the fitted data of the inverter's logic threshold shift in (a) and (b), it is worth to note that the influence of the negative electric field is much more important than that of the temperature. In fact, the shift slopes of the fitted data due to the applied negative field (Fig.3.b) are almost 5 times superior to those due to the temperature (Fig.3.a). Moreover, the shift becomes temperature independent under both stress types when the field exceeds 6.5MV/cm as the ΔVinv slope corresponding to 7MV/cm (see fig.3 (a)) is about 0.04 for DC and 0.01 for AC stress.

As the inverter undergoes under a 0.5 duty-cycle AC NBTI the half stress time of that under DC NBTI, it is supposed that the former induces a shift which equals the half of that induced by the later. However, this has not been observed here (in our experiments) as it is found much less than a half. In fact, the magnitude of the switching logic shift ΔVinv caused by AC stress is representing a 0.25 to 0.46 of the shift caused by DC stress (see fig.4). This ratio is almost the third (for Eox<7MV/cm) and is much less than as expected in the literature and reported elsewhere for the CMOS inverter [6] for a 0.5 duty-cycle AC stress. Besides, this ratio tends to decrease with both temperature and negative field for Eox <7MV/cm. This ratio decrease could be attributed to both the recovery through bias switches, during the half period of the AC stress,

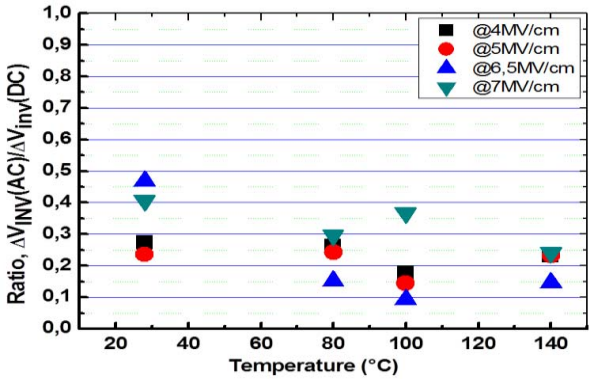


Fig. 4 AC/DC ratio of the inverter`s switching threshold voltage shift.

and temperature acceleration [10]. Therefore, our results reveal that temperature accelerates the recovery as well as it accelerates the stress which explains the AC NBTI–induced shift shrink in (b) in fig 1. This holds true except for the field of 7MV/cm for which the activation energy drastically drops to 0.01 while it is around 0.16 for lower field values (as shown in fig.5). This implies that another physical process is coming out and which could be related to the creation of fixed traps in the oxide and these charge don`t switch or recover after releasing (DC) or alerting (AC) stress. This leads us to think this creation does not have to be related to the pure NBTI degradation and it is only an additional effect due to a highly accelerated stress.

The left shift of the inverter VTC and the decrease of its logic threshold voltage would infer unbalanced input logic states of the inverter by enlarging the voltages range interpreted as logic "1" and shrinking the voltages range interpreted as logic "0" as shown hereafter.

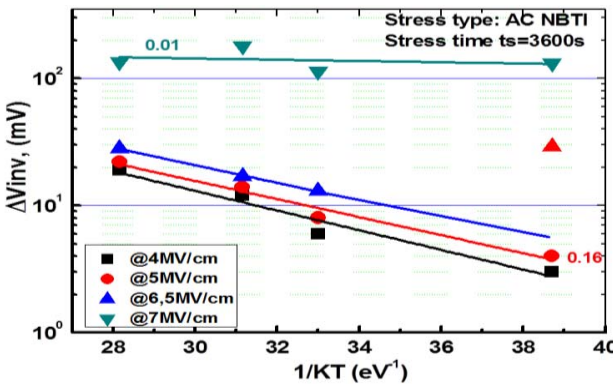


Fig.5 Inverter switching threshold shift due to AC NBTI with two activation energies for fields <7MV/cm and when it equals 7MV/cm.

B. *Logic level voltages under DC& AC NBTI*

Recall that the CMOS inverter logic levels are extracted from the VTC and delimited where the slope equals -1.

Fig.6 shows the extracted output logic voltages under both DC and AC NBTI stresses. It can be clearly seen that the output voltages are weakly impacted by NBTI whatever the applied stress field and temperature. Accordingly, the inverter logic swing, defined as the difference between these two output voltages, remains quasi-constant and thus marginally affected by NBTI.

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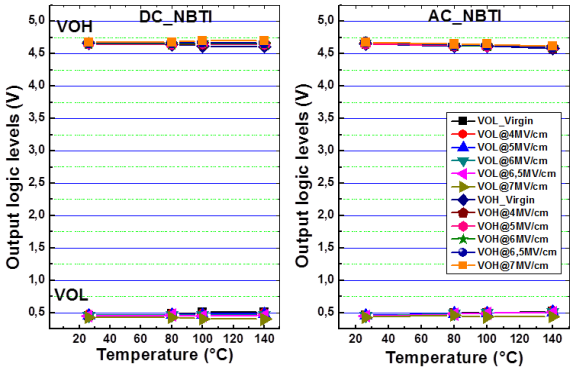

Fig.6 The Inverter output logic levels shift under DC & AC NBTI stresses.

Contrary to the output logic levels, the input logic levels are more affected by NBTI especially at higher negative electric field (see Fig.7). Indeed, both VIL and VIH decreased at higher negative filed (-7MV/cm). However, the shift magnitude is higher under DC NBTI than under AC NBTI. Besides, the shift worsens at elevated temperature under DC NBTI but does not under AC NBTI. The temperature-accelerated recovery can explain this trend under AC NBTI stress since for a given field, the shift decreases with increasing temperature.

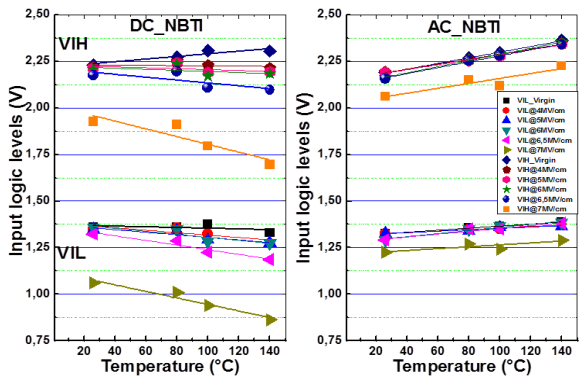

Fig.7 The Inverter input logic levels shift under DC & AC NBTI stresses.

As low and high input voltages have been both affected by NBTI under both DC and AC stress but with different magnitudes, noise margins of the CMOS inverter are thereby expected to be similarly affected. The next section reports the obtained results.

C. *Noise immunity subjected to DC& AC stresses*

The inverter sensitivity to noise is expressed in terms of its noise margins at low and high levels so-called NML and NMH respectively. These margins quantize the size of the legal "0" and "1" logic of the inverter, and set a fixed maximum threshold on the noise value.

Recall that NML and NMH are computed from the extracted logic levels using the following equations:

$$NMH = V_{OH} - V_{IH} \quad (1)$$

$$NML = V_{IL} - V_{OL} \quad (2)$$

Fig. 8 shows the inverter noise margins shift due to DC NBTI (left histograms) and AC NBTI (right histograms). It shows that ΔNMH increases under both stress conditions; we can say that ΔNMH undergoes a positive shift under NBTI whatever the applied stress type. However, the shift magnitude due to DC stress is almost three times larger than that due to AC NBTI stress.

On the other hand, the noise margin low shift trend under AC NBTI differs from that under DC NBTI. It reveals that the noise margin low has decreased under DC NBTI (lower left of fig .8) whereas it has slightly shifted under AC NBTI (lower right of fig.8). Indeed, the shift of noise margin low ΔNML due to AC NBTI is bounded in the range of [-0.06V, 0.07V] presenting thereby almost the fifth of that caused by DC NBTI (up to 0.35V). This is due to the fact that VIL has been less affected by AC- than DC-type NBTI (see fig.7)

It is worth to note that the influence of the temperature on the shift of noise margins under AC-type is quite different from that under DC-type NBTI. While both noise margins high and low shift tend to increase with the applied field and temperature under DC NBTI but this is not always the case under AC NBTI. This is can be explained as follows: during the half period of the AC pulse signal, the circuit is relieved from the field stress and undergoes a recovery aided by

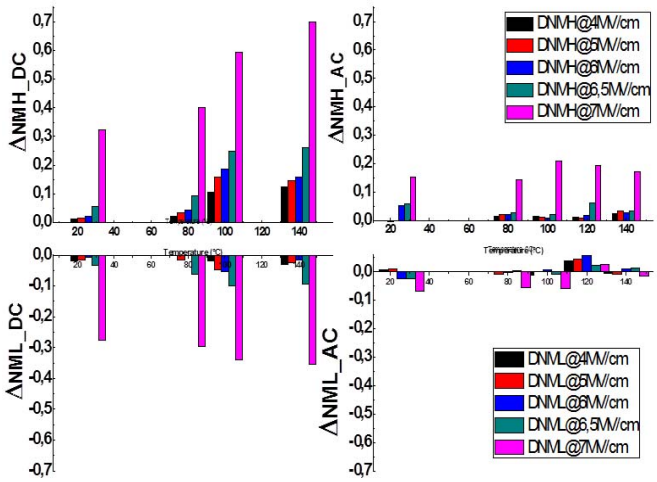

Fig. 8 Noise margin high shift under DC & AC NBTI.

temperature acceleration. This finding is consistent with that reported in [11] and which claims that NBTI can partially and even totally recover with bake. Hopefully, the inverter noise immunity is slightly affected by AC NBTI stress.

IV. Conclusion

In this paper, we have presented an experimental analysis of CMOS inverter DC features trend inder both DC and AC NBTI stresses. The impact of these two types of NBTI on the inverter critical voltages was quantitatively compared. The obtained results have revealed that the CMOS inverter DC features under AC NBTI have undergone a shift much less than the half of that shown under DC NBTI. This shift is seen lower than expected for an AC stress with 50% of duty cycle. Furthermore, the inverter noise immunity and so its robustness is much less affected by AC NBTI than DC NBTI showing a rate of one fifth. These findings are very important for circuit designers to know as DC NBTI ultimately represents the worst case scenario with the highest DC features degradation over time and can lead to an overestimation of the circuit

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degradation and to a pessimistic lifetime prediction and to large design margins introduced to alleviate aging effects.

As our study focuses on 0.5 duty-cycle AC NBTI, a further investigation with different duty cycles remains important to perform in order to more apprehend NBTI degradation dependence on the duty cycle. Therefore a proper understanding of AC NBTI related relaxation issues and adoption of fast measurement method to account for the recovery time will lead to an accurate estimation of the NBTI degradation and therefore to its impact on the lifetime of CMOS circuits.

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REFERENCES

[1] Dieter k. Schroder, "Negative bias temperature instability: What do we understand?," Microelectronics Reliability, Volume. 47, pp. 841–852, 2007.

[2] J.H. Stathis, S. Zafar, "The negative bias temperature instability in MOS devices: A review", Microelectronics Reliability, Volume 46, Issues 2–4, pp. 270-286, 2006.

[3] Vijay Reddy ET all, "Impact of negative bias temperature instability on digital circuit reliability", Microelectronics Reliability, Volume 45, pp. 31–38, 2005.

[4] Bipul. C, Kunhyuk Kang, Haldun Kufluoglu, Muhammad A. Alam, and Kaushik Roy, "Impact of NBTI on the Temporal Performance Degradation of Digital Circuits," IEEE Electron Device Letters, vol. 26, no. 8, pp. 560-562. 2005.

[5] Sang Phill Park, Kaushik Roy & Kunhyuk Kang, "Reliability Implications of Bias-Temperature Instability in Digital ICs," IEEE Design & Test of Computers, vol. 26, no. 6, pp.8-17, 2009.

[6] N. Berbel, R. Fernandez, I. Gil, "Modelling and experimental verification of the impact of negative bias temperature instability on CMOS inverter", Microelectronics Reliability, Volume 49, Issues 9–11, pp.1048-1051, 2009.

[7] S. Khan, S. Hamdioui, "Temperature Dependence of NBTI induced Delay" 16th IEEE International On-Line Testing Symposium (IOLTS), July 5–7, 2010, Corfu Island, Greece, pp. 15-20.

[8] Pobegen. G., Aichinger.T, Nelhiebel. M, , "Dependence of the negative bias temperature instability on the gate oxide thickness", IEEE International Reliability Physics Symposium ,May 2-6 , 2010, Anaheim, CA, USA, pp.1073-1077.

[9] A. Chenouf, B. Djezzar, A. Benabdelmoumene, H.Tahi, "Deep Experimental Investigation of NBTI Impact on CMOS Inverter Reliability", 24th IEEE International Conference on Microelectronics, 17 – 20 December 2012, Algiers, Algeria, pp.1-4.

[10] G. Pobegen, T.Aichinger, M. Neihibel, and T. Grasser, "Understanding Temperature Acceleration for NBTI", IEEE International Electron Devices Meeting, 5-7 December, 2011, Washington DC, USA, pp. 27.3.1- 27-3.4.

[11] Anastasios A. Katsetos, "Negative bias temperature instability (NBTI) recovery with bake", Microelectronics Reliability, Volume 48, Issue 10,pp. 1655-1659, 2008.

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